

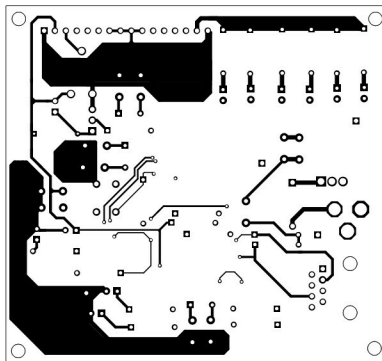


Fig. 6: First layer of the actual-size, double-side PCB

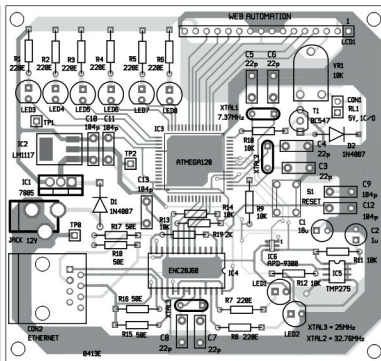


Fig. 8: Component layout for the PCBs in Figs 6 and 7

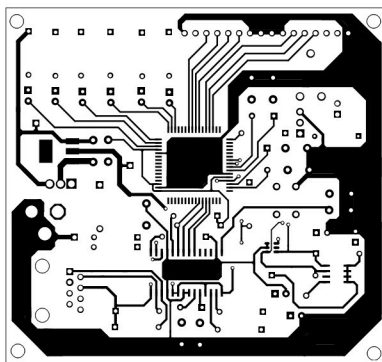


Fig. 7: Second layer of the actual-size, double-side PCB

The source code is compiled and programmed using WinAVR following the below mentioned steps:

1. Install WinAVR
2. Connect the programmer and check the COM port allotted to the device
3. Change the COM port in the make file provided together with the source code
4. Open command prompt and go

to the directory where the program is saved

5. Type 'make clean' and press 'enter'

6. Then type 'make all' and press 'enter'

7. Finally, to program the microcontroller, type 'make program' after connecting the programmer to the target board

Construction and testing

Both the layers of an actual-size,

double-side PCB are shown in Figs 6 and 7 while the component layout is shown in Fig. 8. Assemble the circuit on a PCB as it saves time and minimises assembly errors. Carefully assemble the components and double-check for any overlooked error.

For testing the circuit, refer the test-point table and verify the voltages. To test the functionality of the overall system, proceed as follows:

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download on www.efymag.com website.

1. Connect the system to your computer using Ethernet cable
2. Give your system IP address 10.0.0.xx., where 'xx' should be less than 255
3. Click subnet mask. It will automatically change to 255.0.0.0
4. Change the IP in the source code, say, to 10.0.0.15 (different from the one provided to your computer)
5. Compile and burn the code in the target device
6. Open command prompt and ping 10.0.0.15. If it shows no packet lost, your computer is communicating to the automation system.
7. Now open any Internet browser and write 10.0.0.15 in the url tab. It opens the webpage that is embedded in the source code of the microcontroller with real-time readings from the temperature and light sensors. ●

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